Fast Navigation in a Large Hilbert Space Using Quantum Optimal Control

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The precise engineering of quantum states, a basic prerequisite for the implementation of quantum-enhanced sensing or quantum computing, becomes more challenging with increasing dimension of the control Hilbert space. Optimal pulses in such complex systems can reduce a large number of elementary operations and give access to otherwise elusive states. Here, we use optimal control theory to overcome this problem and drive a single atom in a controllable navigation state of a Rydberg atom. We demonstrate that optimal control becomes increasingly the method of choice when the number of available pulse parameters exceeds the number of relevant dimensions of the Hilbert space. Optimal pulses are found numerically and implemented experimentally with a fidelity of 0.98. Our work opens perspectives for the efficient preparation of complex quantum states, for the implementation of multiqubit gates, and for the optimal control of open quantum systems.

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I. INTRODUCTION

Quantum information science exploits the [illegible] emerging features of light and matter for the [illegible] fields with applications in computing, [illegible] communication [1]. The building blocks of any quantum device are qubits or information carriers, and the ultimate requirement for efficient, controllable and [illegible] their value [2]. The complexity of the concepts is, however, in contrast with the [illegible] when implementing it in an actual physical platform. Both [illegible] and ultimately [illegible] become more and more challenging as precision and the qubit number or the size of the system increases. For example, higher-order terms of the Hamiltonian introduce undesired [illegible] of basic operations [3,4]. The complexity of the system specifies numerical modeling difficult to address [illegible] quantum states on a short time [5,6]. As the [illegible] numerical tools [illegible] which [illegible] in quantum optical control [7,8].

Quantum optimal control [9,10] is a versatile approach to address the challenge of fast navigation in a large Hilbert space. It allows one to design the time-dependent shape of experimental control knobs, electromagnetic fields, for instance, that accomplish a given task as a quantum operation in the best possible way. The starting point is a precise model [illegible], such as the most often [illegible] fidelity, which is [illegible] as a functional of the [illegible] shape of the [illegible] control fields. The success of quantum optimal control [illegible] in maximizing the performance [illegible] while minimizing the use of resources such as [illegible] bandwidth or power [9].

A large number of [illegible] quantum control [illegible] over [illegible] platforms considered for quantum technologies, notably neutral atoms [11], ions [12], photons [13,14], color centers in diamond [15,16] and superconducting qubits [17,18]. Optimized control sequences have [illegible] a single driven qubit and [illegible] a few [illegible] qubits in the [illegible] regime [19–22]. For [illegible] Hilbert spaces, such as the quantum harmonic oscillator [illegible] as an [illegible] of trapped atoms [23,24], optimal control approaches [illegible] to [illegible] conventional quantum engineering designs, providing a [illegible] [25] in resolving [illegible] principle [26]. Here, we go further and demonstrate how quantum optimal control can be used to

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